

CAIE A Level 9702 Physics Paper5 Q1 答题步骤

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1. Defining the problem:

- i. Independent variable.
- ii. Dependent variable.
- iii. Keep constant by

2. Methods of data collection

- i. Take the measurements for the independent variable by appropriate equipment. (涉及到测算的公式要写上去, 例如长度=末端位置-初端位置)
- ii. Repeat the measurements of ... for 3 times and take the average.
- iii. Take the measurements for the dependent variable by appropriate equipment. (涉及到测算的公式要写上去, 例如长度=末端位置-初端位置)
- iv. Repeat the measurements of ... for 3 times and take the average.
- v. Change the for different independent variable and repeat the above procedure to get a group of data.

3. Method of analysis

- i. 原公式的变形过程——线性关系
- ii. 根据公式变形处理数据 (平方、取对数...).
- iii. Plot a graph of ... against ...
- iv. The relationship valid if a straight (not) passing through the origin produced. 如果知道, 标上截距斜率的正负。
- v. 计算过程+...= 一个含 gradient 或者 y-intercept 的式子。

4. Additional details:

i. **Precaution:**

- (1) 有电路(有磁感线圈): wearing heat-proof and insulating gloves when touching the equipment to avoid injury because of high temperature and shock.
- (2) Switch the current off when it is not in use.
- (3) Use a protective resistor to reduce current.
- (4) 电路在水里: ensure that the electrical connections are waterproof.
- (5) 有热源或发热的物体: wearing heat-proof gloves when touching the equipment to avoid injury because of high temperature.
- (6) Goggles to prevent splashes from hot water
- (7) 防止实验器材掉落地面摔坏: use cushion/foam/sandbox for.... Use a sand tray under heavy weights and make sure weights don't fall on your foot.
- (8) 防止桌上 (旋转) 的实验器材飞出去: use safety screen.
- (9) 气压: safety precaution relating to pressure, e.g. safety screen.
- (10) 有 (强) 光源的实验: dark glasses/shield lamps to avoid eyes damage.

ii. **加大强度/功率来减小误差(to reduce uncertainty):**

- (1) 有光源: dark room, high intensity lamp, light source is placed close to ...
- (2) 造磁场: large current or large number of wire turns in coils to produce large magnetic field.

*其他有关磁场的点: Magnetism

Keep Hall probe at right angles (perpendicular) to magnetic field by fixing to rule.

Calibrate Hall probe in a known magnetic field. Repeat experiment with Hall probe reversed and average.

Avoid external alternating magnetic fields.

Use large current and large number of turns.

Safety: use heat-proof gloves to position coil.

iii. 其他:

- (1) 测水中深度: use a high container to give a wide range of depth in water.
- (2) 有电流: DC/AC
- (3) 找圆心: determine the position of the center by finding the intersection of two diameters.
- (4) 保持水平: spirit level/保持垂直: spirit level, vertical rule, correctly positioned set square indicated at right angles to the horizontal surface, plumb line shown in appropriate position.
- (5) 慢速增加 (转速、功率等) increase the ... gradually.
- (6) 稳定后再测量: take measurements of ... until steady state.
- (7) 测周期、频率或其他时间相关参量: details of using stroboscope/video camera and play back in slow motion.
 - For measuring time period of oscillations, find time for 10 oscillations and then divide.
 - Use fiducial markers to time oscillating objects.
- (8) 固定器材: iron standing/ clamp to maintain the equipment at a constant position.
- (9) 做标记
- (10) 减小摩擦: use air cushion track, clean surfaces of ... to ensure that the surfaces are smooth.
- (11) 均匀加热: use magnetic stirrers to make sure that the solution is heated evenly.
- (12) For experiments with light weights or wind, close windows & switch off air conditioning to avoid draughts.
- (13) In an experiment with object being dropped, make sure object released with no/constant velocity. Can use electromagnets or a spring loaded device.
- (14) For falling objects, use a guide to keep motion in correct direction.
- (15) 发射具有恒定初速度的物体: use the same spring with a constant spring constant, and compress the spring at a constant strain (by using caliper to measure the strain every time) to release the object.
- (16) 排水法测体积或质量或物体沾水: use hairdryer to dry the object.
- (17) 使用量角器: correctly positioned protractor to measure θ , e.g. protractor with its centre over... and its straight edge parallel to...

附: 实验仪器:

Measuring Instruments

Physical quantity	Instruments	
Distance	Meter rule Vernier caliper Micrometer	Use set square/ spirit level to make sure apparatus perpendicular repeat measurements in different positions and average
Time	Stop watch Light gate	- o Measure distance between light gates - o Connect light gates to time loggers - o Calculate time of fall by using data from loggers – time difference between when the first and second beam are broken
Velocity	Light gate	Measure length of the object and calculate velocity $v=L/t$
Mass	Balance	
Volume	Measuring cylinder	
Angle	Protractor	
Current	Ammeter galvanometer	
Voltage	Voltmeter	
Resistance	Ammeter + Voltmeter Ohmmeter	
Temperature	Thermometer	
Magnetic field strength	Hall probe	
Time period(sound)	CRO microphone	o Adjust time-base and y-gain to achieve a suitable waveform
Amplitude(sound)		o Calculate amplitude by finding height in terms of boxes on grid of waveform and multiplying by y-gain o Calculate period by counting boxes of grid occupied by a full waveform and multiply by time-base setting
Wavelength(light)	Diffraction grating	
Intensity	LDR	

Supportive instruments: Steel retort stand, clamp, string